

OM SHAH

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PROFESSIONAL SUMMARY

Cybersecurity graduate (Bachelor, UTS, June 2026) targeting junior SOC Analyst, Security Analyst, and vulnerability management roles. Delivered Python automation in a production migration environment, reducing case handling time by 35%. Built red-team simulations including DNS exfiltration via DLL sideloading on fully patched Windows 11, then built blue-team detection logic for the same traffic patterns. CompTIA Security+ in progress.

TECHNICAL SKILLS

Security Testing	Nmap / NSE, Metasploit, Wireshark, mitmproxy, Burp Suite, vulnerability assessment, penetration testing, web application testing
Defensive Security	Threat modelling (STRIDE), SIEM concepts, incident detection, log and traffic analysis, SOC monitoring fundamentals, risk assessment, information security management
Frameworks	MITRE ATT&CK, OWASP Top 10, NIST, CVE / NVD, ISO 27001
Programming	Python, C, Selenium, HTML / CSS, JavaScript, Git / GitHub, JSON / CSV, Bash
Platforms	Kali Linux, VMware, VirtualBox, AWS (IAM, Security Groups), Cisco Packet Tracer, Jira, Postman

CERTIFICATIONS

CompTIA Security+ – in progress (expected 2026)

PROFESSIONAL EXPERIENCE

Migrate Zone <i>Automation Engineer · Part-time</i>	Dec 2024 – Present
- Reduced average case handling time by 35% (8.0h to 5.3h) by engineering modular Python automation for document validation, compliance triage, risk flagging, and case routing. - Consolidated financial audit checks into a single workflow, eliminating duplicate processing and manual cross-referencing across immigration case files. - Built an Excel prefill pipeline that mapped validated case data into lodgement templates, eliminating repetitive entry and reducing compliance errors. - Replaced manual policy monitoring with a web scraper that tracked government updates and surfaced regulatory changes in real time.	
Migrate Zone <i>Junior Automation Engineer · Part-time</i>	Jul 2023 – Nov 2024
- Co-built a PDF collation and OCR pipeline that caught dozens of missing, expired, or lapsed documents each month before lodgement, reducing compliance risk. - Automated case data entry with Python scripts that pushed validated fields into visa workflow systems, replacing hours of repetitive manual input. - Implemented scan-quality gates using OCR coverage and confidence scoring to flag unreadable pages before reviewer handoff. - Authored internal SOP documentation that enabled non-technical staff to run the automation toolkit without developer support.	
ITonKey <i>Python Backend Developer Intern · Part-time</i>	Aug 2024 – Oct 2024
Shipped a Python invoicing system that generated branded, multi-page PDF invoices from structured business data, replacing a manual client process.	

- Implemented stock-tracking and loss-reporting features in an inventory platform, improving operational visibility and reducing manual reconciliation.

PROJECTS

Sentinel – Automated Network Vulnerability Scanner

UTS Capstone, 2025

- Built an automated network scanner that mapped discovered devices and services to known CVEs from the National Vulnerability Database, reducing manual assessment from hours to minutes.
- Improved detection accuracy by handling SMB, RDP, and edge-case service fingerprints with CPE-to-CVE matching.
- Generated risk-prioritised JSON reports with per-finding verification commands for rapid security-team triage.

Python, Nmap / NSE, NVD CVE API, JSON · Video demo and development journal at omshahinfo.com

Covert Channel – DNS Exfiltration via DLL Sideloads

Independent, 2025

- Exfiltrated a file from a fully patched Windows 11 endpoint with zero alerts from Defender, AMSI, SmartScreen, or Windows Firewall using a native C implant sideloaded through a Microsoft-signed binary.
- Executed five MITRE ATT&CK techniques end-to-end: HTML smuggling (T1566.002), MOTW bypass (T1553.005), DLL sideloading (T1574.002), native code execution (T1027.009), and DNS tunnelling (T1048.003).
- Built a paired blue-team detector that flagged tunnelling via Shannon entropy, burst-frequency analysis, and subdomain-length anomalies in an isolated VMware lab.

C, Python, DNS (RFC 1035), Wireshark, VMware · Interactive PCAP visualisation at omshahinfo.com

SMS Bulk-Style Classifier (Machine Learning)

UTS, 2026

- Trained a 1D CNN with GloVe embeddings to score SMS messages for bulk/automated wording, outperforming Naive Bayes and SVM baselines on precision, recall, and F1.
- Deployed the model end-to-end via FastAPI on Hugging Face Spaces with an interactive browser demo for live inference.

Python, TensorFlow, GloVe, FastAPI · Live demo and technical report at omshahinfo.com

EDUCATION

Bachelor of Cybersecurity

2023 – 2026

University of Technology Sydney

Key subjects: Cybersecurity, System Security, Digital Forensics, Cryptography, Cloud Security (AWS), Information Security Management, Routing and Switching, Data Analytics